PROJECT REPORT

RescueFlow AI

Intelligent traffic management with emergency vehicle priority

YOLOv8 vehicle detection • Ambulance tracking • Adaptive signals

CURRENT IMPLEMENTATION

A real-time Flask dashboard combining YOLOv8, ByteTrack, lane analytics, fail-safe emergency preemption, video streaming, and optional MySQL reporting.

Prepared from the implemented RescueFlow AI codebase

27 JULY 2026

Executive Overview

THE PROBLEM

Fixed-duration traffic lights do not adapt well to changing vehicle queues. They also delay ambulances when emergency priority depends on manual intervention. RescueFlow AI addresses both issues with camera-based traffic intelligence and a controlled signal-preemption workflow.

THE SOLUTION

- Detect common road vehicles using the standard YOLOv8 nano model.
- Track vehicle identities and estimate lane-wise counts, queue length, and waiting time.
- Detect ambulances with a dedicated custom YOLO checkpoint and maintain persistent ByteTrack IDs.
- Safely transfer green priority using yellow clearing and all-red safety intervals.

4	5	3	8 / 8
TRAFFIC LANES	VEHICLE CLASSES	SIGNAL STATES	TESTS PASSING

DESIGN PRINCIPLE

Emergency detection is intentionally independent of the normal traffic detector. If the custom ambulance weights are missing or unavailable, ordinary vehicle analysis and the dashboard continue to operate instead of failing.

Objectives & Capabilities

TRAFFIC INTELLIGENCE

- Process recorded traffic video or a live webcam stream.
- Recognize cars, buses, trucks, motorcycles, and bicycles.
- Assign stable IDs to reduce double counting across frames.
- Divide the camera view into four logical traffic lanes.
- Calculate current and unique counts for every lane.
- Estimate queue distance and average waiting time.

EMERGENCY & OPERATIONS

- Select the busiest lane for adaptive green priority.
- Use 20, 35, or 50-second green windows based on density.
- Confirm ambulance detections over multiple frames.
- Track approach, crossing, clearing, and brief detection loss.
- Expose live state through JSON endpoints and a responsive dashboard.
- Optionally store snapshots and emergency events in MySQL.

EXPECTED IMPACT

Adaptive
DECISION STYLE

Fail-safe
EMERGENCY MODE

Real time
VISIBILITY

System Architecture



Traffic Intelligence Workflow

01

Capture

Read the next frame from video or webcam.

02

Detect

YOLOv8n identifies supported road-vehicle classes.

03

Track

Centroid association assigns persistent vehicle IDs.

04

Measure

Counts, queues, wait estimates, and density are calculated.

05

Optimize

The busiest lane receives a density-based green window.

06

Publish

Overlays, dashboard state, and optional database logs update.

DENSITY-BASED TIMING

LOW 20s

MEDIUM 35s

HIGH 50s

Emergency Vehicle Priority

The dedicated ambulance model runs independently and uses ByteTrack IDs. A track must be observed across three frames before signal priority is activated, reducing the risk of a one-frame false trigger.



TRACK LIFECYCLE

- APPROACHING — before the configured entry line.
- CROSSING — between entry and exit lines.
- CLEARED — past the exit line; priority can end.
- LOST — absent beyond the missed-frame tolerance.

SAFETY BEHAVIOUR

- No instant green transfer across conflicting lanes.
- Lane changes restart the safe clearing sequence.
- Detection loss triggers all-red recovery.
- Normal operation resumes automatically.

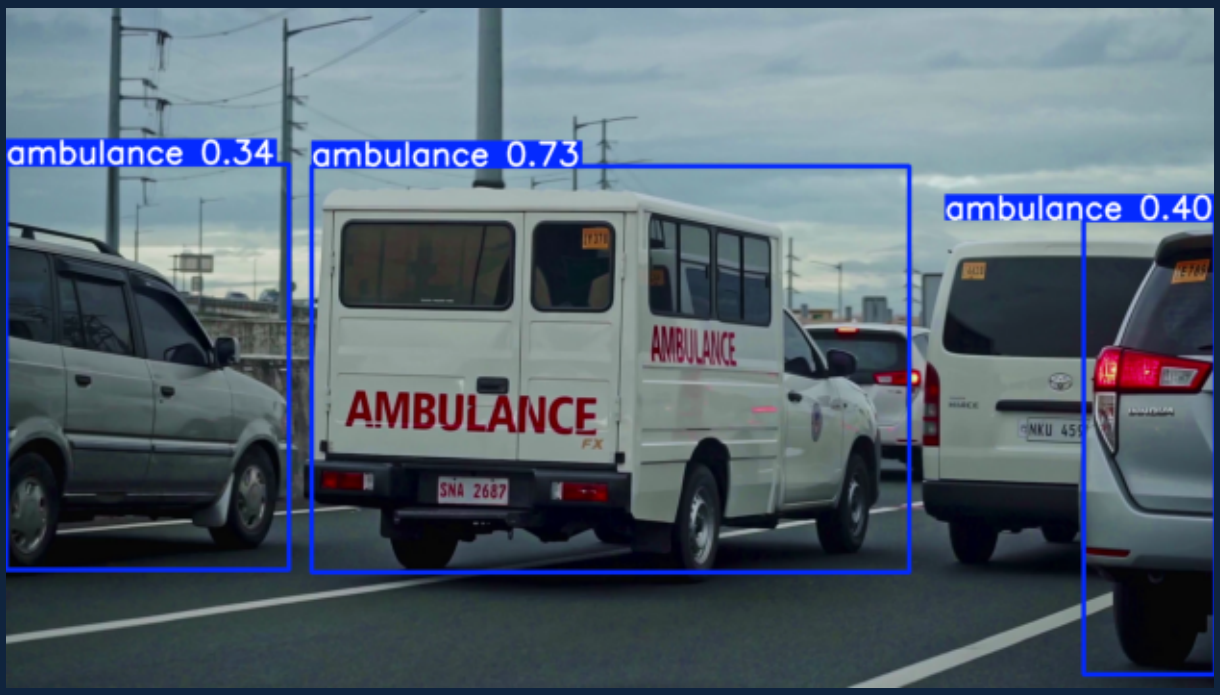
CONFIGURED DETECTION ZONE

ENTRY 35% DIRECTION DOWN EXIT 82%

Ambulance Model Deployment

1	3.0M	640	Ambulance
MODEL CLASSES	PARAMETERS	INPUT SIZE	CLASS

INFERENCE SAMPLE FROM BUNDLED AMBULANCE VIDEO



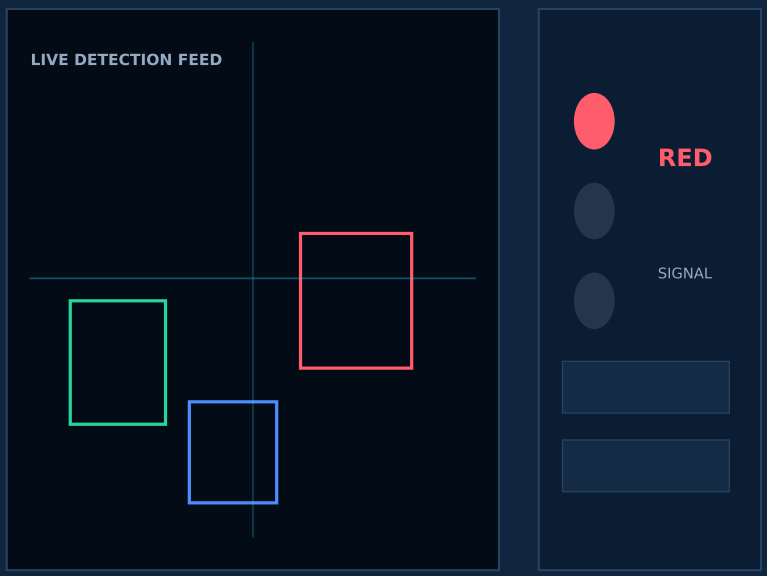
Selected frame: 3 detection(s), highest confidence 72.6% at a 0.25 display threshold.

DEPLOYMENT PATH

Training output: best.pt → **assets/models/ambulance.pt**

Operations Dashboard

LIVE MONITORING SURFACE



Responsive dark command-center UI with signal glow, emergency status, countdown, lane bars, vehicle-class totals, and storage health.

LIVE ENDPOINTS

GET /

Dashboard

GET /video_feed

MJPEG stream

GET /stats

Live state

GET /reports

Daily summary

STATE EXPOSED TO THE UI

- vehicle totals
- unique IDs
- density
- green lane
- signal status
- countdown
- lane queues
- vehicle classes
- emergency identity
- confidence
- tracking status
- priority phase
- detector health
- database health
- daily report metrics

Technology Stack & Setup

COMPUTER VISION

OpenCV 4.11

AI DETECTION

Ultralytics YOLO 8.3

TRACKING

ByteTrack + centroid

BACKEND

Flask 3.1

FRONTEND

HTML / CSS / JavaScript

STORAGE

MySQL (optional)

RUNNING THE APPLICATION

```
01  python3 -m venv venv
02  source venv/bin/activate
03  pip install -r requirements.txt
04  python3 main.py --path assets/videos/ambulance.mp4
05  Open http://127.0.0.1:5000
```

The ambulance model is loaded automatically from assets/models/ambulance.pt.

Verification & Conclusion

AUTOMATED VERIFICATION — 8 TESTS PASSED

- Missing ambulance weights do not break normal operation
- Three-frame confirmation and entry/exit lifecycle
- Brief detection loss tolerance
- ByteTrack assignment compatibility fallback
- Yellow → all-red → emergency-green → recovery sequence
- Direct priority when the emergency lane is already green
- Safe recovery when an emergency track disappears
- Safe clearing when emergency priority changes lanes

CURRENT RESULT

RescueFlow AI is an integrated prototype with live vehicle detection, lane analytics, adaptive signal selection, ambulance tracking, safe preemption, a responsive operations dashboard, and optional persistence.

RECOMMENDED NEXT STEPS

- Record model precision, recall, mAP50, and mAP50–95.
- Calibrate camera-specific lane and crossing lines.
- Test with day, night, rain, and occlusion scenarios.
- Measure end-to-end latency on deployment hardware.

CONCLUSION

The project demonstrates how computer vision can improve ordinary traffic flow while giving emergency vehicles a safer and more predictable path through intersections. Its modular design allows the custom detector, signal policy, database, and user interface to evolve independently.